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Test-Time Speculation

Continuously Adapting the Draft Model During Generation Sustains Speculative Speedups Throughout Long Chain-of-Thought Outputs

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Speculative decoding accelerates LLM inference by having a small draft model propose candidate tokens that a large target model verifies in parallel. In practice, however, the acceptance length of trained speculators—including state-of-the-art systems EAGLE-3, DFlash, and PARD—decays to near 1 after a few thousand output tokens, eliminating the speedup for long chain-of-thought generation. Test-Time Speculation (TTS) fixes this by adapting the draft model online, using the target-model evaluations that the verification step already performs, at zero added inference cost. On Qwen-3, Qwen-3.5, and Llama 3.1 model families, TTS improves acceptance length by up to 72% and 41% on average over the best static speculator, translating to 2.1–3.4× wall-clock speedup on 5K-token generation tasks.

AT A GLANCE

Problem: Trained speculators are distilled offline on short sequences; their acceptance length decays toward 1 on the long CoT outputs that matter most for reasoning tasks.

Why it matters: Speculative decoding is the leading inference acceleration technique, but its speedup evaporates precisely for the long-horizon tasks that consume the most compute.

Method: Online KL distillation of the draft model using target-model token probabilities already computed by the verification step, plus a proximity regularizer to prevent forgetting.

Result: Up to 72% improvement in acceptance length; 2.1–3.4× wall-clock speedup on 5K+ token outputs.

Evidence: MATH-500, HumanEval+, long-document summarization; Qwen-3/3.5 and Llama 3.1 families; ablations on regularizer and step count.

The Big Picture

Speculative decoding exploits the asymmetry between generation and verification: a small, fast draft model proposes k tokens, and the large target model verifies all k in one forward pass—accepting the longest prefix consistent with its distribution. When the draft model agrees with the target model, the effective throughput multiplies by k; when it does not, the cost is at most one wasted draft pass.

The technique's Achilles' heel is draft quality at long context lengths. All existing speculators are trained on datasets of typical generation lengths (hundreds to low thousands of tokens); at inference, reasoning-intensive tasks such as MATH, LiveCodeBench, and multi-step QA generate five to ten thousand tokens. The draft model's internal representations, tuned for short contexts, drift from the target's as the context grows, and acceptance length collapses.

Test-Time Speculation reframes speculative decoding as a continual learning problem: the draft model is a student that should track the target's distribution throughout generation, not just at the start.

Why Existing Approaches Fall Short

Static draft models (EAGLE-3, DFlash, PARD): All trained offline on sequences far shorter than inference-time CoT outputs. No mechanism exists to update the draft as the context lengthens.

Longer training sequences: Simply including longer sequences in offline training is expensive, requires representative CoT data that may not be available, and still does not adapt to the specific prompt or intermediate reasoning steps of a given inference call.

Online re-training approaches: Prior online adaptation methods for NLP require periodic re-training runs with additional target-model calls, breaking the generation pipeline and negating the speedup.

Core Mechanism

At each speculative decoding step, the target model already evaluates the k draft tokens and produces per-token probability distributions $\pi^*(\cdot \mid x_{<j})$ for $j = t, \ldots, t+k$. These distributions are the exact student-teacher signal needed to distill the draft model for the current context—and conventional speculative decoding discards them after acceptance/rejection decisions.

TTS re-uses these distributions to take one gradient step on the draft model after each verification pass. The gradient step minimizes a KL distillation loss plus a proximity regularizer that prevents the draft from forgetting its base distribution. Because the gradient is taken on a small draft model (1–3B parameters) and no additional target-model calls are needed, the per-step overhead is negligible.

Technical Formulation

Let π^* denote the target model and π_θ the draft model with parameters θ. At step t, the token-level KL distillation loss over the k verified positions is:

$$\mathcal{L}_t(\theta) = \frac{1}{k} \sum_{j=t}^{t+k} D_{\mathrm{KL}}(\pi^*(\cdot \mid x_{<j}) \parallel \pi_\theta(\cdot \mid x_{<j}))$$

The proximity regularizer prevents catastrophic forgetting:

$$\Omega(\theta, \theta_{t-1}) = \|\theta - \theta_{t-1}\|_2^2$$

The combined update is a single gradient step:

$$\theta_t = \theta_{t-1} - \eta \nabla_\theta [\mathcal{L}_t(\theta_{t-1}) + \beta \Omega(\theta_{t-1}, \theta_{t-2})]$$

with learning rate η following a cosine decay schedule over the generation horizon, and regularization weight β set so that total parameter shift remains within a small budget.

Learning or Inference Procedure

Standard speculative decoding (baseline): Frozen draft model; acceptance length decays with context length.

TTS (proposed):

1. Initialize π_θ from any pre-trained speculator.
2. At each verification step:
   (a) Accept/reject draft tokens as usual.
   (b) Compute $\mathcal{L}_t$ using the target distributions already obtained.
   (c) Take one gradient step to update θ.
3. Continue generation with the updated draft model.

No additional target-model calls. The gradient step on the small draft model adds $< 5\%$ latency overhead in practice.

What the Guarantee Says

The distillation loss minimized at each step is the expected log-probability ratio between the target and draft at the *current* context. Minimizing it pushes the draft toward the target's distribution at the positions just verified. The proximity regularizer bounds the per-step parameter shift, ensuring that adaptation at step t does not degrade draft quality at step $t-1$'s context distribution beyond a controlled amount. Together, these properties imply that the acceptance rate converges to a level bounded by the target's entropy at the current context—the theoretical maximum for any draft model—rather than decaying toward 1.

Experimental Findings

- **MATH-500 (long mathematical reasoning):** TTS achieves 3.4× wall-clock speedup over the target at 5K tokens; the best static speculator achieves 1.5× at the same length.
- **HumanEval+ (long code generation):** 2.7× vs. 1.8× for TTS and the best static speculator respectively.
- **Acceptance length trajectory:** Static speculators show a monotonic decay from ~ 4.5 tokens at 500 tokens to ~ 1.2 at 5K tokens; TTS maintains ~ 4.0 tokens throughout.
- **Model families:** Gains are consistent across Qwen-3-8B, Qwen-3.5-32B, and Llama-3.1-70B with their respective draft models.
- **Compatibility:** TTS improves acceptance length for all three base speculators (EAGLE-3, DFlash, PARD); EAGLE-3 + TTS achieves the best absolute acceptance length.

Ablations and Interpretation

No proximity regularizer: The draft overfits to the current context window and loses generalization to the next step; acceptance length drops by 18% compared to full TTS.

Multiple gradient steps per verification: Two steps per round adds compute without improving acceptance; one step is both necessary and sufficient.

Fixed vs. cosine learning rate: Cosine decay improves stability and final acceptance length by $\sim 5\%$; a fixed high rate causes oscillation in later generation steps.

Gains grow with output length: Improvement over the base speculator is near-zero at < 500 tokens (where the training distribution covers the context) and reaches its maximum at $> 5{,}000$ tokens, confirming that TTS specifically addresses the out-of-distribution long-context problem.

Reference

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